Problem Pack: Exponent Laws and Perfect Squares from Mathematics 8 • Unit 2 Review

Question 1a

Problem: 1. Simplify each expression. Your final answer should contain only positive exponents.
a) x^4 x^7

Student Work:

x^{11}

Feedback

Question 1b

Problem: b) a^{12} ÷ a^5

Student Work:

a^7

Feedback

Question 1c

Problem: c) (m³)⁴

Student Work:

m^{12}

Feedback

Question 1d

Problem: d) (2pq²)³

Student Work:

2³p³q^6

Feedback

Question 1e

Problem: e) (3x^2 / 2y)^2

Student Work:

9x^2 ÷ 4y^2

Feedback

In question 1e, you went wrong when squaring the x^2 term on the top of the fraction. You forgot to apply the exponent outside the parentheses to the exponent on the x, thinking it stayed as x^2 instead of multiplying the powers. Try rewriting the numerator as (3²)(x)2)^2 and apply the power of a power rule again.

Question 1f

Problem: f) 7r^0

Student Work:

7r

Feedback

In question 1f, you went wrong in the first step by forgetting the rule that any non-zero number raised to the power of zero equals one. You treated the r^0 as just r instead of turning it into a 1. Remember what happens to any base when its exponent is zero, and try multiplying that value by the 7 again.

Question 2a

Problem: 2. Simplify completely. Use any exponent laws you need, and leave only positive exponents. a) [(2m^3n2)^2(4mn3)] / (8m^2n)

Student Work:

4m^6n4 x 4mn^3 ÷ 8m^2n = 2m^5n6

Feedback

Question 2b

Problem: b) (3x^2y)3 / (9x^4y2)

Student Work:

27x^6y3 ÷ 9x^4y2 = 3x^2y

Feedback

Question 3

Problem: 3. Classify each number as a perfect square or non-perfect square. If it is a perfect square, give its principal square root. 225 30 4/9 49/20 0.16 0.6

Student Work:

√225 = 15 PS NPS PS √4/9 = 2/3 NPS √0.16 = 0.4 PS NPS

Feedback

In question 3, you went wrong on the second half of the list by mixing up which numbers are perfect squares and which are non-perfect squares. Remember that a fraction is a perfect square if both its top and bottom numbers are square numbers, and decimals are perfect squares if they can be made by multiplying a terminating decimal by itself. Look closely at the fraction and decimal options that you labeled as non-perfect squares and check if they have a clean square root.

Question 4a

Problem: 4. State how many real square roots each number has. Briefly explain. a) 81

Student Work:
2 sr they are 9 and −9

Feedback

Question 4b
Problem: b) 0

Student Work:
1 sr It is 0

Feedback

Question 4c
Problem: c) −49

Student Work:
0 sr

Feedback

Question 5
Problem: 5. Evaluate √49. Then list all solutions of x^2 = 49. Explain why the answers are different.

Student Work:
√49 =

Feedback
In question 5, you went wrong by combining the two different parts into one answer and missing the explanation of why they are different. A radical sign only asks for the principal or positive square root, while the equation asks for all numbers that can multiply by themselves to reach the target. Try separating your answer into the single positive value for the radical and both the positive and negative values for the equation.